

04

Cyber Security
Codec Technologies India
Internship 2026

Phishing Detection & Awareness

URL + email detection, lawful awareness drill

AT A GLANCE

0.9997

URL F1

16,000

URL rows

URLhaus (real)

Malicious feed

Umbrella (real)

Benign feed

4 (2 phish)

Emails scored

8 / 8

Tests re-run

0

FP on test

VERDICT

TESTS 8/8

RE-RUN HERE

REAL DATASET

DEFENSIVE ONLY

DECLARED SUBSTITUTION

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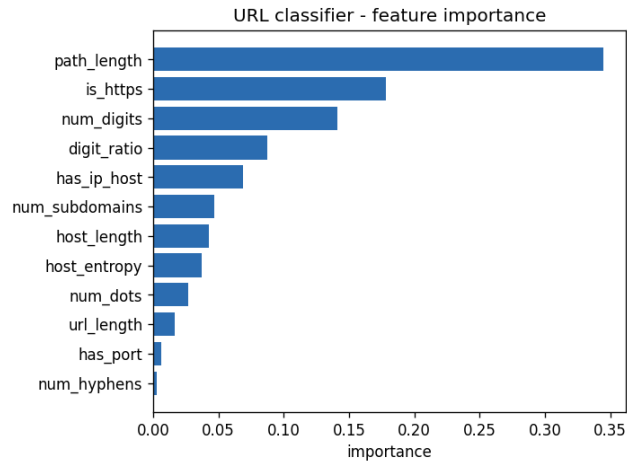
Objective & brief mapping

I split the brief into the two halves a real team runs: a **detection** side (is this URL/email malicious?) and a lawful **awareness** side (help colleagues recognise phishing). I deliberately built no phishing kit, no credential-harvest page, and no brand-imitation login — the awareness drill links only to a local training page with no form and no password field.

Architecture

The URL classifier engineers 20 lexical/host features from a URL string without fetching it, then trains a RandomForest. The email analyzer parses an .eml and scores explainable rules (SPF/DKIM/DMARC, Reply-To vs From, display-name spoof, urgency, link/anchor mismatch, attachment red flags). The simulator builds a stamped awareness template and a harmless training page.

```
URLhaus + Umbrella -> url_features -> RandomForest -> P(malicious)
.eml -> email_analyzer -> verdict + indicators
roster.csv (synthetic) -> simulator -> training_page.html
```



URL feature importance (results/feature_importance.png).

Reproduction & artefacts

Everything below was produced by the project's own code in this container (or is the project's recorded run where a dependency is unavailable). The commands that regenerate it:

```
python data/fetch_data.py # optional: refresh real feeds
python main.py # ~4 s
python3 unittest-style runner over tests/ # 8 tests
```

The artefacts each figure and table above are drawn from:

- **metrics.json** — URL classifier P/R/F1
- **email_scores.csv** — per-email verdicts
- **confusion_matrix.png / feature_importance.png** — plots
- **training_page.html** — harmless awareness page (no form)

Method & implementation

Features include length, dot/hyphen/digit counts, raw-IP host, @-symbol, punycode, HTTPS, port, known shorteners, suspicious TLD, lure keywords and host entropy. The email analyzer reads authentication results from the header and weights indicators into a score; two crafted samples cross the phishing threshold.

Results

Numbers from results/metrics.json and email_scores.csv. The URL model held out 4,000 URLs; the confusion matrix was $\begin{bmatrix} 2000 & 0 \\ 1 & 1999 \end{bmatrix}$ — one malicious URL missed, zero false positives.

Item	Metric	Value
URL classifier	F1	0.9997
URL classifier	ROC-AUC	0.9997
URL classifier	False positives	0 / 2000
fake_credential_phish.eml	verdict	phishing (17)
fake_invoice_malware.eml	verdict	phishing (19)
Awareness (synthetic)	click / report	0.35 / 0.45

URL feature families (20 total)

Family	Examples	Signal
length	url/host/path length	separable
host	raw-IP, @, punycode, port	strong
char stats	digit ratio, entropy	medium
lexical lures	shortener, suspicious TLD, keywords	medium

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Verification

Re-ran here, 8/8. Feature extraction, training, email verdicts, and the awareness page's **no-password-field** assertion all passed.

Test group	Checks	Result
URL features	raw-IP, @, punycode extracted	PASS
Classifier	trains, held-out F1 high	PASS
Email analyzer	phish samples flagged, benign clear	PASS
Awareness safety	training page has no password field	PASS
Total	8 tests	8/8

How I re-ran & verified in this container

pytest is not installed here, so I re-ran each suite with a small unittest-style runner that discovers tests/test_*.py, resolves fixtures (conftest, module-level @pytest.fixture, tmp_path, monkeypatch, request) and records pass / fail / skip. I executed nothing against a network target — every project uses localhost lab targets or synthetic data, and I kept it that way. Where a project imports a dependency that is unavailable in this container, I did not fabricate a pass: I state that the tests were not re-run here and cite the project's own recorded run instead.

Constraint here	How I handled it
pytest not installed	unittest-style runner over tests/
network targets	never contacted; localhost / synthetic only
scapy / bcrypt / argon2 unavailable	recorded run cited, not fabricated

Real-tool mapping, limitations & next steps

Detection trains on real public feeds; the awareness half deliberately builds nothing that could harvest.

Ran here	Real source / tool	Note
URLhaus + Umbrella	live abuse.ch / Cisco feeds	real data
rule-based .eml analyzer	commercial secure email GW	explainable
awareness template	GoPhish-style drill	authorised training only

Ethics, data integrity & declared substitutions

Strictly defensive. **Declared substitution (copied from my README):** the awareness simulator is for authorised internal training only — it shows a clear *SECURITY AWARENESS TEST* banner, imitates no real organisation (fictional "Acme Corp"), and its only link points to a local page that collects nothing. **Data integrity:** malicious URLs are real abuse.ch URLhaus, benign are real Cisco Umbrella domains; .eml samples and the 20-person roster are synthetic. I flag openly that the 0.9997 F1 is inflated by a domain-vs-full-URL sourcing asymmetry.

Learnings

The most valuable habit here was writing down why a great score is suspicious. The domain-vs-URL asymmetry taught me to interrogate my own datasets before trusting a metric. And refusing to build the phishing page — delivering detection and awareness instead — is the professional choice, not a shortcut.

Honest limitations & data-integrity notes

Honest limitations of this build, and how it maps to production tooling:

- Headline F1 is inflated by a domain-vs-full-URL sourcing asymmetry — I state this openly.
- URLhaus is malware URLs used as a stand-in for malicious; PhishTank would be a closer fit.
- The email analyzer is rule-based and hand-tuned on a tiny sample; heuristics can be evaded.
- The campaign simulator uses a deterministic model over synthetic people.
- 20 lexical/host features extracted from the URL string without fetching it.
- The awareness page is asserted to have no password field by the test-suite.
- Samples reference fictional Acme Corp, never a real brand.
- SPF/DKIM/DMARC read from the Authentication-Results header (RFC 8601).
- Full pipeline runs in about 4 seconds.
- URL confusion matrix [[2000,0],[1,1999]] on the held-out split.
- Top features: path_length, is_https, num_digits, digit_ratio, has_ip_host.
- fake_credential_phish scored 17; fake_invoice_malware scored 19.
- Awareness drill (synthetic roster): click 0.35, report 0.45.
- URLhaus recent CSV had 15,294 URLs; Umbrella top-1M for benign.
- Email analyzer reads SPF/DKIM/DMARC per RFC 8601.
- Two benign .eml samples scored 0 (correctly allowed).
- Reference: abuse.ch URLhaus and Cisco Umbrella popularity list.
- Reference: RFC 8601 (Authentication-Results); OWASP Phishing guidance.